

Романов Александр Дмитриевич
 241-321 - Практикум 2
 № 1 (1,3)

$$1) \begin{vmatrix} 3 & 5 \\ 2 & 4 \end{vmatrix} = 3 \cdot 4 - 5 \cdot 2 = 2$$

$$3) \begin{vmatrix} x-y & x+y \\ x+y & x-y \end{vmatrix} = (x-y)^2 - (x+y)^2 =$$

$$= \underline{x^2 - 2xy + y^2} - \underline{x^2 + 2xy + y^2} = -4xy$$

№ 2

$$1) \begin{vmatrix} 1 & 3 & 2 \\ 2 & -1 & 3 \\ -3 & 2 & 1 \end{vmatrix} = -1 + 8 - 27 - 6 - 6 - 6 =$$

$$= -38$$

$$2) \begin{vmatrix} 3 & 4 & -5 \\ 8 & 7 & -2 \\ 2 & -1 & 2 \end{vmatrix} = \overset{168}{3 \cdot 7 \cdot 2} - 16 + 40 + 70 - 6 - \overset{256}{4 \cdot 6 \cdot 4} =$$

$$= 272 - 378 = -106$$

$$3) \downarrow$$

✓

$$\begin{vmatrix} 2 & 1 & 3 \\ 0 & 4 & 2 \\ -1 & 2 & 3 \end{vmatrix} = 24 - 2 + 0 + 12 - 8 - 0 =$$

$$= 26$$

~ 3

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} \quad M_{11} = \begin{vmatrix} 5 & 4 \\ 2 & 1 \end{vmatrix} = -3$$

$$M_{21} = \begin{vmatrix} 2 & 3 \\ 2 & 1 \end{vmatrix} = -4$$

$$A_{13} = M_{13} = \begin{vmatrix} 4 & 5 \\ 3 & 2 \end{vmatrix} = -7$$

$$A_{32} = -M_{32} = -\begin{vmatrix} 1 & 3 \\ 4 & 4 \end{vmatrix} = 8$$

$\sim 4 (1, 4)$

$$1) \begin{vmatrix} 1 & 0 & -1 \\ 2 & 3 & 4 \\ -1 & 2 & 1 \end{vmatrix}$$

$$a) \begin{vmatrix} 1 & 0 & -1 \\ 2 & 3 & 4 \\ -1 & 2 & 1 \end{vmatrix} = 1 \begin{vmatrix} 3 & 4 \\ 2 & 1 \end{vmatrix} + 0 - 1 \begin{vmatrix} 2 & 3 \\ -1 & 2 \end{vmatrix} =$$

$$= 3 - 8 - 4 - 3 = -12$$

$$d) \begin{vmatrix} 1 & 0 & -1 \\ 2 & 3 & 4 \\ -1 & 2 & 1 \end{vmatrix} = 0 + 3 \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} - 2 \begin{vmatrix} 1 & -1 \\ 2 & 4 \end{vmatrix} =$$

$$= 3 - 3 - 12 = -12$$

$$4) \begin{vmatrix} i & j & k \\ 1 & -2 & 4 \\ 3 & 0 & 2 \end{vmatrix} =$$

$$a) \begin{vmatrix} i & j & k \\ 1 & -2 & 4 \\ 3 & 0 & 2 \end{vmatrix} = i \cdot \begin{vmatrix} -2 & 4 \\ 0 & 2 \end{vmatrix} - j \begin{vmatrix} 1 & 4 \\ 3 & 2 \end{vmatrix} + k \begin{vmatrix} 1 & -2 \\ 3 & 0 \end{vmatrix} =$$

$$= -4i + 10j + 6k$$

$$d) \begin{vmatrix} i & j & k \\ 1 & -2 & 4 \\ 3 & 0 & 2 \end{vmatrix} = -j \begin{vmatrix} 1 & 4 \\ 3 & 2 \end{vmatrix} - 2 \begin{vmatrix} i & k \\ 3 & 2 \end{vmatrix} + 0 =$$

$$= 10j - 2(2i - 3k) = -4i + 10j + 6k$$

$$\sim 5(1, 3)$$

f) ↓

$$\begin{vmatrix} x & y & x+y \\ 2 & 1 & -1 \\ 1 & 3 & 2 \end{vmatrix} = x \begin{vmatrix} 1 & -1 \\ 3 & 2 \end{vmatrix} - y \begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix} + (x+y) \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} =$$

$$= 5x - 5y + (x+y)(6-1) = 10x$$

$$3) \begin{vmatrix} a & b & c \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix} = a \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} - b \begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} + c \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} =$$

$$= a + 6b - 3c$$

$$1) \begin{vmatrix} 0 & 1 & -1 & 2 \\ -1 & 2 & 1 & 0 \\ 3 & 2 & -1 & 3 \\ 1 & 6 & 1 & 1 \end{vmatrix} \xrightarrow{8(1,2)} \begin{vmatrix} 0 & 1 & -1 & 2 \\ 0 & 8 & 2 & 1 \\ 0 & -16 & -4 & 0 \\ 1 & 6 & 1 & 1 \end{vmatrix} \begin{matrix} \sim \\ \text{II} + \text{IV} \\ \text{IV} - 3\text{IV} \end{matrix}$$

$$= -1 \cdot \begin{vmatrix} 1 & -1 & 2 \\ 8 & 2 & 1 \\ -16 & -4 & 0 \end{vmatrix} = -(0 + 16 - 64 + 64 + 4 + 0) =$$

$$= -20$$

2) ↓

$$\begin{vmatrix} 1 & a & 2 & 4 \\ 2 & b & 1 & 3 \\ 3 & c & 5 & -4 \\ 4 & d & -2 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 3 & 1 & 3 & 1 \\ 0 & 2 & 1 & 3 \\ -3 & -4 & 2 & 4 \\ 6 & 2 & 5 & 2 \end{vmatrix} \begin{array}{l} \text{III} + \text{I} \\ \text{IV} - 2\text{I} \end{array}$$

$$\sim \begin{vmatrix} 3 & 1 & 3 & 1 \\ 0 & 2 & 1 & 3 \\ 0 & -3 & 5 & 5 \\ 0 & 0 & -1 & 0 \end{vmatrix} = 3 \begin{vmatrix} 2 & 1 & 3 \\ -3 & 5 & 5 \\ 0 & -1 & 0 \end{vmatrix} =$$

$$= 3(0 + 0 + 9 - 0 - 0 + 10) = 3 \cdot 19 = 57$$

Романов Александр Дмитриевич.

241-321

Практикум 2

в. 2 (2)

$$2) \begin{vmatrix} 1 & -2 & 3 & 4 \\ 1 & -1 & 2 & 3 \\ 1 & 2 & 5 & 3 \\ 1 & 3 & 2 & -2 \end{vmatrix} \begin{matrix} \\ \text{II}-\text{I} \\ \text{III}-\text{I} \\ \text{IV}-\text{I} \end{matrix} = \begin{vmatrix} 1 & -2 & 3 & 4 \\ 0 & 1 & -1 & -1 \\ 0 & 4 & 2 & -1 \\ 0 & 5 & 1 & -6 \end{vmatrix} =$$

$$= 1 \cdot \begin{vmatrix} 1 & -1 & -1 \\ 4 & 2 & -1 \\ 5 & 1 & -6 \end{vmatrix} = -12 + 5 - 4 + 10 + 1 - 24 =$$
$$\underline{\underline{-24}}$$